

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Crankshaft locking pin, Part Number 3163020
- Gear puller, Part Number 3163069
- Loctite® 609, Part Number 3823718
- Loctite® 242, Part Number 3824040 or 3824041

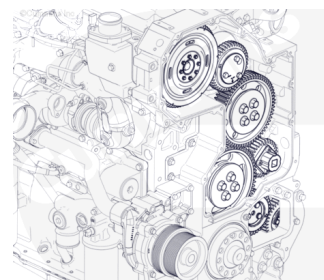
Additional Service Items

- Shims
- Pry bar
- ENVIROSOL™ 655 solvent
- Camshaft timing wedge
- Lubriplate™

General Information

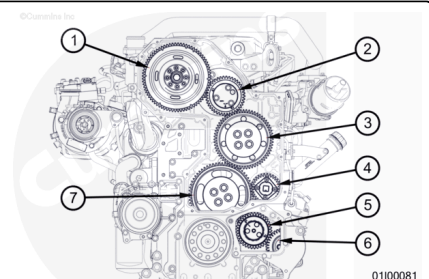
Front Gear Assembly Configuration

This procedure describes the removal, inspection, and installation of the gear train.



This illustration depicts the gear train with a solid-style lower idler gear.

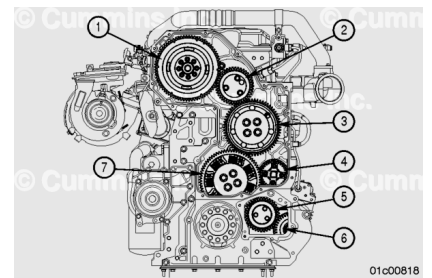
1. Camshaft gear, valve
2. Camshaft idler gear, adjustable
3. Idler gear, upper
4. Accessory gear
5. Idler gear, fuel pump, adjustable



6. Fuel pump gear
7. Idler gear, lower solid.

This illustration depicts the gear train with a scissor-style lower idler gear.

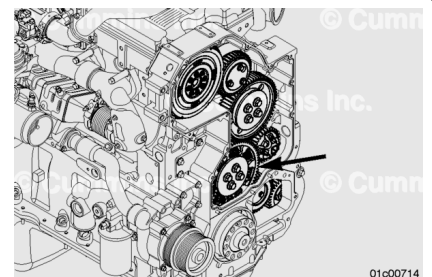
1. Camshaft gear, valve
2. Camshaft idler gear, adjustable
3. Idler gear, upper
4. Accessory gear
5. Idler gear, fuel pump, fixed/doweled
6. Fuel pump gear
7. Idler gear, lower scissor.



Scissor Gear Definitions

⚠ CAUTION ⚠

Do not attempt to remove any gears before reading scissor gear definitions. Serious personal injury can result if instructions are not followed.



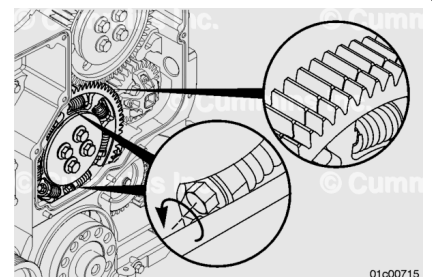
⚠ CAUTION ⚠

Do not attempt to remove any gears before reading scissor gear definitions. Engine damage can result if instructions are not followed.

The following terms describe the conditions of the scissor gears for removal, installation, and operation.

Unloaded

- The gear will be unloaded when removing, installing, and setting gear backlash.
- Unload the gear by backing out two gear adjusting screws until the gear teeth align.
- The idler scissor gear will be loaded after the gear backlash is set.



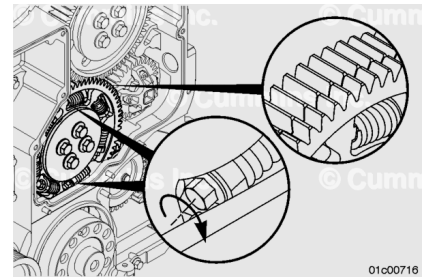
Loaded

- The gear teeth are loaded (functional) and are spread between the mating gear teeth.

- The gear is loaded by tightening all of the adjusting screws.

Torque Value: 29 n·m [21 ft-lb]

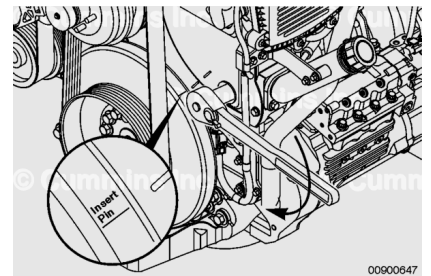
- The gear teeth will be spread and no movement will be felt against the crankshaft gear. The gear will be in the loaded condition for engine operation.



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ CAUTION ⚠

The timing pin and wedge must be installed before any gears are removed and must remain in place until all gears are fully installed and tightened. Failure to install the timing pin and wedge can lead to improper timing of the engine, resulting in severe engine damage or engine failure.

- Disconnect the batteries. See equipment manufacturer service information.

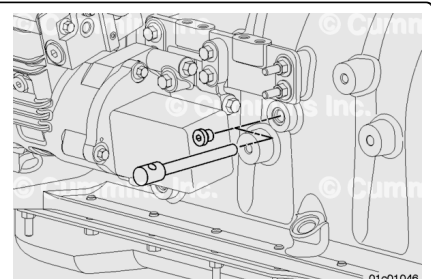
Rotate the crankshaft **clockwise** until the "insert pin" mark on the outside diameter of the engine vibration damper is aligned with the mark on the gear housing.

Remove the plug from the timing pin boss.

Insert crankshaft locking pin, Part Number 3163020, into the timing pin boss hole.

The green band on the timing pin will be lined up with the surface of the timing pin boss of the block.

The pin is **not** correctly seated in the crank notch if the green band is either completely visible outside the block or is **not** visible at all.



The crankshaft will be locked in place when the pin is properly seated.

- Remove the exhaust gas recirculation (EGR) crossover tube. Refer to Procedure 011-070 in Section 11. (</qs3/pubsys2/xml/en/procedures/377/377-011-070-tr.html>)
- Remove the rocker lever cover and gasket. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/377/377-003-011-tr.html>)
- Remove the naturally aspirated air compressor plumbing and components, if equipped.

The timing code is located on the engine dataplate.

© Cummins Inc. © Cummins Inc.		
Engine Cert Id.	Displacement	15L CPL
	Fuel Rate at Rated Power	
Timing T.D.C. (inj/valve)	6	Idle Speed (RPM)
Valve Lash Cold (in/mm)		Int.

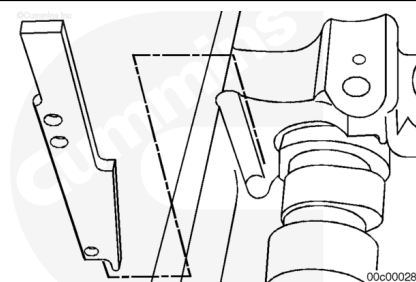
01c00720

⚠ CAUTION ⚠

Do not drive the wedge into place. Engine damage can occur.

Insert the correct timing wedge to lock the valve camshaft. Lightly tap the top of the timing wedge with a mallet to confirm it is properly seated.

Verify the wedge with the proper angle is used and the wedge is properly seated in its installed position.



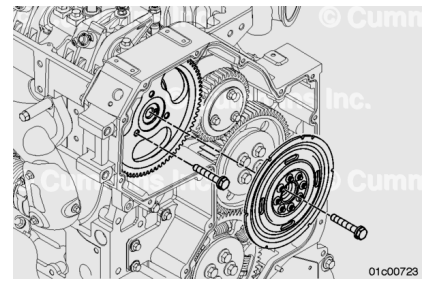
- Remove the camshaft position sensor. Refer to Procedure 019-363 in Section 19. (</qs3/pubsys2/xml/en/procedures/377/377-019-363.html>)
- Remove the upper gear cover. Refer to Procedure 001-079 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-079.html>)
- Remove the crankshaft position sensor. Refer to Procedure 019-365 in Section 19. (</qs3/pubsys2/xml/en/procedures/377/377-019-365.html>)
- Remove the lower gear cover, if needed. Refer to Procedure 001-080 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-080.html>)

Remove

Valve Camshaft Gear

⚠ CAUTION ⚠

Do not attempt to remove any gears before reading scissor gear definitions. Serious personal injury can result if instructions are not followed.



⚠ CAUTION ⚠

Do not attempt to remove any gears before reading scissor gear definitions. Engine damage can result if instructions are not followed.

The lower idler scissor gear, if equipped, **must** be unloaded during removal of the valve camshaft gear. To unload the gear, two of the gear adjusting screws **must** be backed out far enough to allow the teeth on the scissor gear to be aligned.

Unload the lower idler scissor gear.

Note : This step is not needed for a gear train with a solid lower idler gear.

Remove the valve camshaft gear mounting capscrew and the camshaft tone wheel.

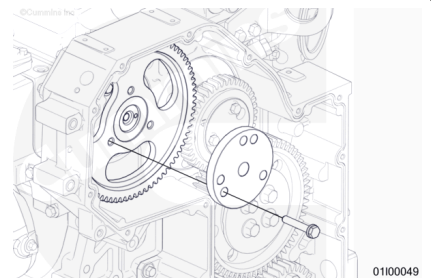
Note : Do not disassemble the camshaft tone wheel.

Install the gear puller, Part Number 3163069.

Install the valve camshaft gear mounting capscrew completely.

Loosen the capscrew two turns.

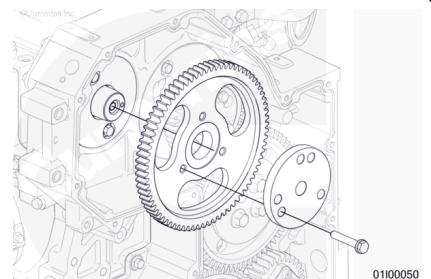
Use the gear puller to remove the valve camshaft gear from the camshaft nose.



Remove the gear puller.

Do **not** completely remove the camshaft gear mounting capscrew when the three-jaw puller is used. The capscrew serves as a safety catch by preventing the gear from slipping off of the camshaft's tapered nose.

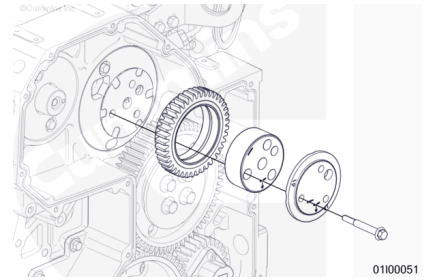
Remove the mounting capscrew, spacer, and valve camshaft gear.



Camshaft Idler Gear (Adjustable)

⚠ CAUTION ⚠

Do not attempt to remove any gears before reading scissor gear definitions. Serious personal injury can result if instructions are not followed.



⚠ CAUTION ⚠

Do not attempt to remove any gears before reading scissor gear definitions. Engine damage can result if instructions are not followed.

⚠ CAUTION ⚠

The camshaft idler gear shaft can fall out and be damaged when the camshaft idler gear is removed.

The lower idler scissor gear, if equipped, **must** be unloaded during the removal of any gears. To unload the lower idler scissor gear, two lower idler scissor gear adjusting screws **must** be backed out far enough to allow the teeth on the lower idler scissor gear to be aligned.

Unload the lower idler scissor gear.

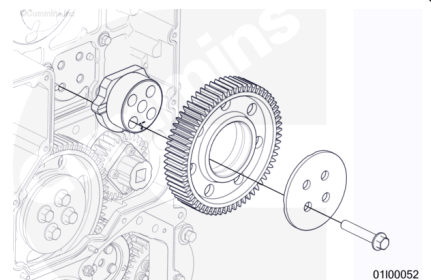
Note : This step is not needed for a gear train with a solid lower idler gear.

Remove the retainer capscrews, retainer, camshaft idler gear, and idler gear shaft.

Upper Idler Gear

⚠ CAUTION ⚠

Do not attempt to remove any gears before reading scissor gear definitions. Serious personal injury can result if instructions are not followed.



⚠ CAUTION ⚠

Do not attempt to remove any gears before reading scissor gear definitions. Engine damage can result if instructions are not followed.

The lower idler scissor gear, if equipped, **must** be unloaded during the removal of any gears. To unload the lower idler scissor gear, two lower idler scissor gear adjusting screws **must** be backed out far enough to allow the teeth on the lower idler scissor gear to be aligned.

Unload the lower idler scissor gear.

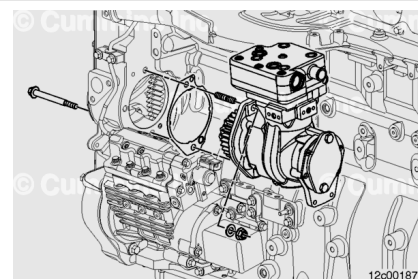
Note : This step is not needed for a gear train with a solid lower idler solid gear.

Remove the retainer capscrews, retainer, upper idler gear, and spacer/shaft assembly.

Accessory Drive Gear

If equipped, remove the air compressor. Refer to Procedure 012-014 in Section 12.

(/qs3/pubsys2/xml/en/procedures/377/377-012-014.html)



Fuel Pump Idler Gear

⚠ CAUTION ⚠

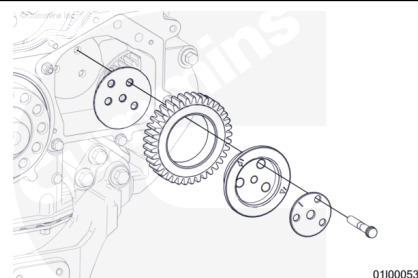
Do not attempt to remove any gears before reading scissor gear definitions. Serious personal injury can result if instructions are not followed.

⚠ CAUTION ⚠

Do not attempt to remove any gears before reading scissor gear definitions. Engine damage can result if instructions are not followed.

⚠ CAUTION ⚠

The fuel pump idler gear shaft can fall out and be damaged when the fuel pump idler gear is removed.



The lower idler scissor gear, if equipped, **must** be unloaded during the removal of any gears. To unload the lower idler scissor gear, two lower idler scissor gear adjusting screws

must be backed out far enough to allow the teeth on the lower idler scissor gear to be aligned.

Unload the lower idler scissor gear.

Note : This step is not needed for a gear train with a solid lower idler solid gear.

Remove the capscrews, retaining plate, shaft, fuel pump idler gear, and thrust plate.

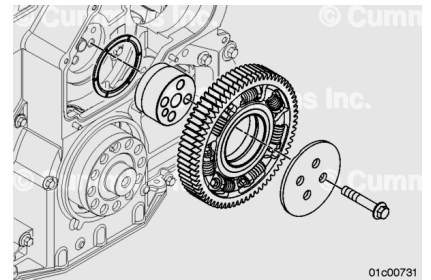
Lower Idler Scissor Gear

⚠ CAUTION ⚠

Do not attempt to remove any gears before reading scissor gear definitions. Serious personal injury can result if instructions are not followed.

⚠ CAUTION ⚠

Do not attempt to remove any gears before reading scissor gear definitions. Engine damage can result if instructions are not followed.



The lower idler scissor gear **must** be unloaded during removal of any gears. To unload the gear, two of the gear adjusting screws **must** be backed out far enough to allow the teeth on the scissor gear to be aligned.

Unload the lower idler scissor gear.

Remove the idler scissor gear capscrews.

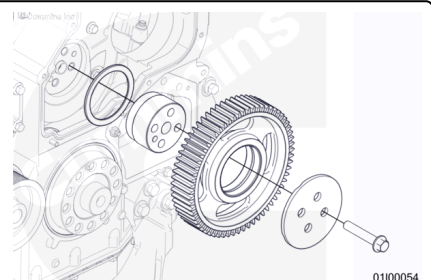
Remove the retainer, lower idler scissor gear, shaft, and thrust bearing.

Lower Idler Solid Gear

Remove the lower idler gear capscrews.

Remove the retainer plate, lower idler gear, shaft, and thrust bearing.

Note the slots in the gear retainer plate face toward the gear.

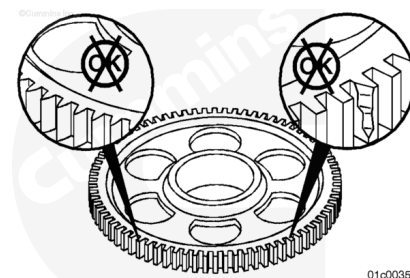


Inspect for Reuse

Inspect the inner diameter bushings for heavy wear or pitting. For bushings with a clinch joint on the inner diameter, inspect the connection for any separation or delamination that exposes underlying steel.

Inspect the gear teeth on all gears for pitting, heavy wear, or breakage.

Replace the suspect gear. Inspect the mating gear for resulting damage.



Inspect the idler shafts for heavy wear.

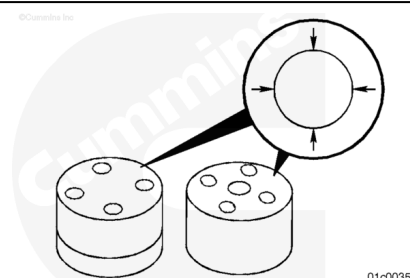
Measure the idler shafts outside diameters.

Lower Idler Shaft Outside Diameter		
mm		in
88.880	MIN	3.4992
88.894	MAX	3.4998

Upper Idler Shaft Outside Diameter		
mm		in
88.880	MIN	3.4992
88.894	MAX	3.4998

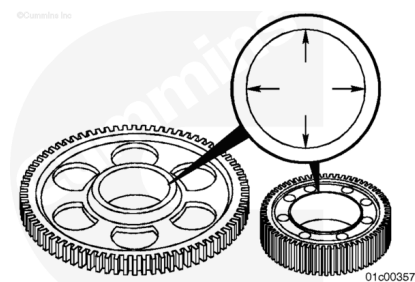
Camshaft Idler Gear Shaft Outside Diameter		
mm		in
95.343	MIN	3.7537
95.357	MAX	3.7542

Fuel Pump Idler Shaft Outside Diameter		
mm		in
78.344	MIN	3.0844
78.358	MAX	3.0850



Inspect the idler gear bores for heavy wear or pitting.

Measure the inside diameter of the lower idler gear, upper idler gear, camshaft idler gear, and fuel pump idler gear bores.



Lower Idler Gear Bore Inside Diameter

mm		in
88.963	MIN	3.5025
88.989	MAX	3.5035

Upper Idler Gear Bore Inside Diameter

mm		in
88.963	MIN	3.5025
88.989	MAX	3.5035

Camshaft Idler Gear Bore Inside Diameter

mm		in
95.426	MIN	3.7569
95.452	MAX	3.7580

Fuel Pump Idler Gear Bore Inside Diameter

mm		in
78.426	MIN	3.0876
78.452	MAX	3.0887

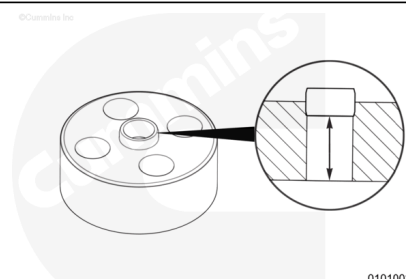
Inspect the thrust bearing(s) and thrust bearing plates for pitting, cracking, scratching, uneven wear, or other damage. Replace the thrust bearing(s) or thrust bearing plates. Inspect the corresponding gear if damage is found.

Inspect capscrews for corrosion and signs of deformation. Replace capscrews if damage is found.

Inspect capscrews for threadlocker residue. Clean with a wire brush.

Inspect the dowel ring installation depth into the idler gear idler shaft. Verify the dowel ring is installed correctly by taking a depth measurement from the outside face of the idler shaft to the top of the dowel ring.

Installation depth readings should be between 28.89 to 29.73 mm [1.13 to 1.17 in].

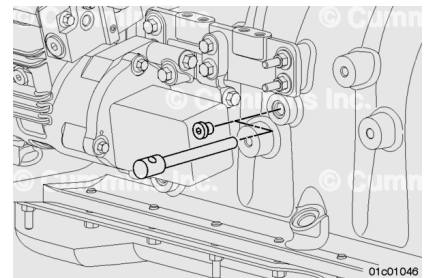


Use the following procedure if the dowel installation depth is out of range. Refer to Procedure 001-111 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-111.html>)

Install

⚠ CAUTION ⚠

The timing pin and wedge must be installed before any gears are removed and must remain in place until all gears are fully installed and tightened. Failure to install the timing pin and wedge can lead to improper timing of the engine, resulting in severe engine damage or engine failure.



⚠ CAUTION ⚠

The gear train must be assembled in the same sequence as this procedure lists. Failure to follow this sequence may result in severe engine damage.

If **only** one gear is removed and installed:

- It is necessary to measure gear backlash after installing gear.
 - Gear backlash **must** be measured to each gear the subject gear touches. Refer to Procedure 001-055 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-055.html>)

If the valve camshaft gear, camshaft idler gear, or fuel pump gear is removed and installed:

- It is necessary to adjust gear backlash after installing gears.

If **only** the fuel pump gear is removed and installed:

- Measurement or adjustment of backlash is **not** needed.

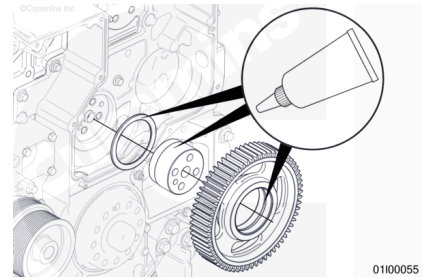
If removing more than one gear:

- Installation order is critical. See ordered steps below.
- Adjust gear backlash after installing gears.
- Perform a rotation check.

Lower Idler Solid Gear

⚠ CAUTION ⚠

The lower idler gear thrust bearing plate and shaft must be oriented such that the oil drain from the head is not blocked. If blocked, engine operation will be affected.



Apply Lubriplate™ to the thrust bearing, shaft, and lower idler gear.

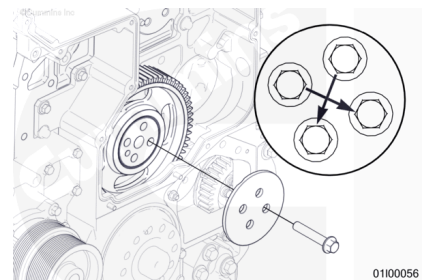
Install the thrust bearing plate, shaft, and lower idler gear. The shaft **must** be installed with the word “TOP” (stamped on the end of the shaft) toward the top of the engine.

Verify the hole in the thrust bearing plate, shaft, and retainer plate are in line with the oil drilling in the block.

The slots in the gear retainer plate face toward the gear.

Install the retainer plate and capscrews.

Tighten all capscrews in the alternating pattern shown in the illustration.



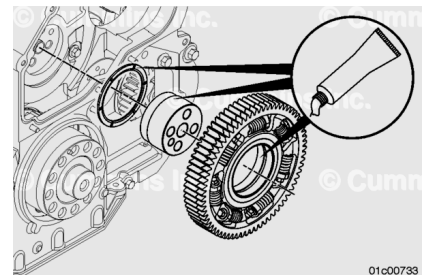
Torque Value:

1. 30 n•m [22 ft-lb]
2. Rotate each capscrew an additional 60 degrees in the same alternating patterns.

Lower Idler Scissor Gear

⚠ CAUTION ⚠

Do not attempt to remove any gears before reading scissor gear definitions. Serious personal injury can result if instructions are not followed.



⚠ CAUTION ⚠

Do not attempt to remove any gears before reading scissor gear definitions. Engine damage can result if instructions are not followed.

The lower scissor idler gear **must** be unloaded during installation. To unload the gear, two of the gear adjusting screws **must** be backed out far enough to allow the teeth on the scissor gear to be aligned.

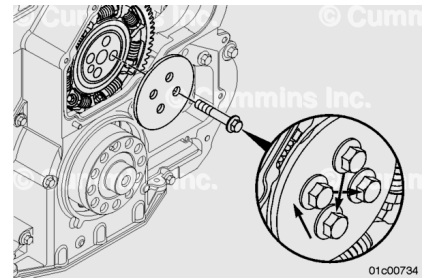
Apply Lubriplate™ to the thrust bearing, shaft, and lower scissor idler gear.

Install the thrust bearing, shaft, and lower scissor idler gear.
The shaft **must** be installed with the word "TOP" (stamped on the end of the shaft) toward the top of the engine.

The slots in the gear retainer plate face toward the scissor gear.

Install the retainer and capscrews.

Tighten all capscrews in the alternating pattern shown in the illustration.



Torque Value:

1. 30 N·m [22 ft-lb]
2. Rotate each capscrew an additional 60 degrees in the same alternating patterns.

Note : The lower idler scissor gear **must not** be loaded until all other gears are installed and tightened.

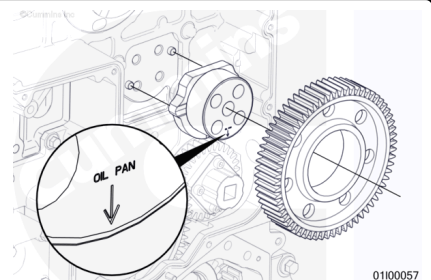
Upper Idler Gear

Apply Lubriplate™ to the thrust bearing plate, shaft, and upper idler gear.

The spacer plate/shaft assembly is doweled to the block with two standard dowels.

Install the mounting spacer plate/shaft assembly and the upper idler gear.

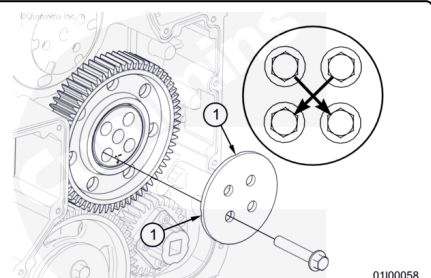
The spacer plate/shaft assembly **must** be installed with the word "OIL PAN", stamped on the front face of the shaft, oriented toward the oil pan.



The slots (1) in the gear retainer plate face toward the gear.

Install the retainer and capscrews.

Tighten all capscrews in the alternating pattern as shown in the illustration.



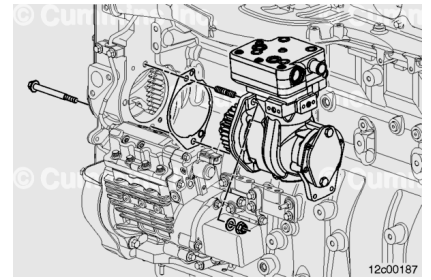
Torque Value:

1. 30 N·m [22 ft-lb]
2. Rotate each capscrew an additional 60 degrees in the same alternating patterns.

Accessory Drive Gear

If equipped, install the air compressor. Refer to Procedure 012-014 in Section 12.

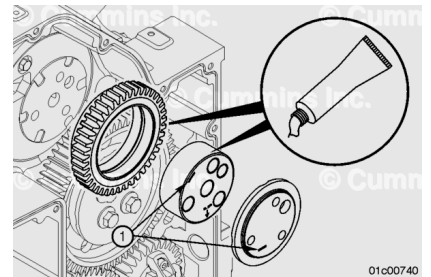
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Camshaft Idler Gear (Adjustable)

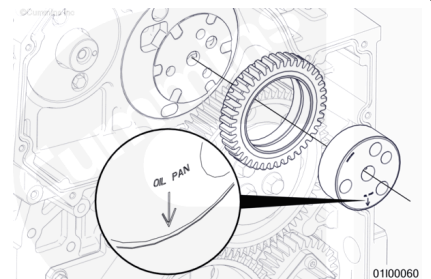
⚠ CAUTION ⚠

The camshaft idler gear retainer and shaft must be oriented with the part number facing outward, such that the oil drain from the head is not blocked. If blocked, engine operation will be affected.



Apply Lubriplate™ to the shaft and camshaft idler gear.

Install the shaft and the adjustable camshaft idler gear. The shaft **must** be installed with the words "OIL PAN" and the arrow (stamped on the front face of the shaft) pointing toward the oil pan.

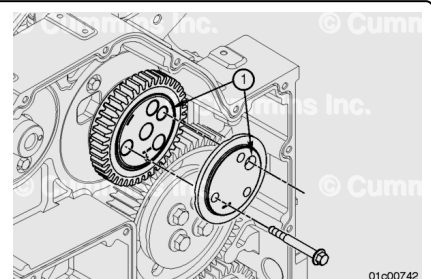


The slots in the gear retainer plate face toward the gear.

Install the retainer and capscrews. Hand tighten the capscrews and then loosen by one "flat," which is 60 degrees.

Verify the oil hole in the retainer plate and shaft (1) is in line with the oil drilling in the head.

Note : The camshaft idler gear is installed loosely. It is moved into its final position later in the timing process.



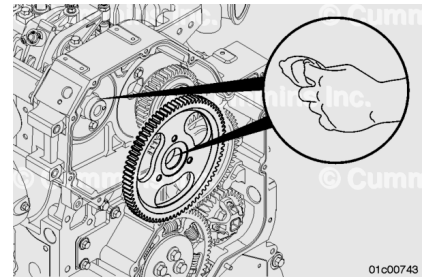
Valve Camshaft Gear

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ CAUTION ⚠

The valve camshaft nose and valve camshaft gear must be clean and dry prior to assembly. Wipe off the shaft surface and gear with a lint-free cloth and do not touch the surfaces after wiping. Engine damage can occur.



⚠ CAUTION ⚠

The camshaft gear retaining capscrew torque process must be completed within fifteen (15) minutes of Loctite® application or severe engine damage may occur.

Scrape off any Loctite® residue from the bore of the valve camshaft gear.

Clean the valve camshaft and camshaft gear bore with Envirosol™ 655 solvent, or equivalent, and a lint-free cloth.

Do **not** spray solvent directly onto the camshaft nose. Spray solvent onto a clean, lint-free cloth, and then wipe the camshaft nose clean. No oil residue can remain and these surfaces are **not** to be touched once cleaned.

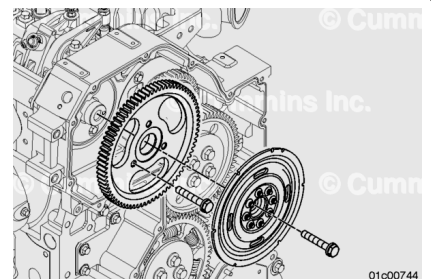
Apply a thin coat of Loctite® 609, Part Number 3823718, to the camshaft nose and camshaft gear bore.

Install the valve camshaft gear loosely on the camshaft nose taper.

The valve camshaft gear needs to be fully engaged on the camshaft nose taper, yet loose enough so it can rotate on the camshaft nose.

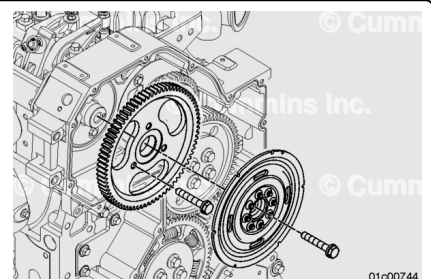
Install the tone wheel and capscrew.

Verify that the key on the tone wheel matches up properly with the keyed hole in the camshaft.



Hand tighten the capscrew and then loosen by one "flat," which is 60 degrees.

Note : The valve camshaft gear is still loose at this point to allow the adjustable idler gear to be properly positioned. Do not seat the gear.

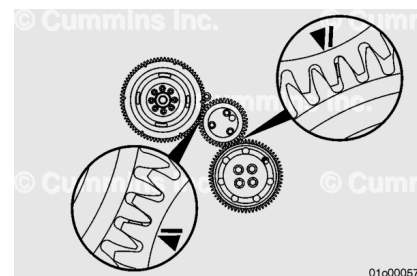


Adjust

Camshaft Idler Gear (Adjustable)

Insert, as illustrated, a narrow shim (0.1016 mm [0.004 in]) into the gear teeth mesh between the camshaft idler gear and the upper idler gear, as well as a narrow shim (0.1016 mm [0.004 in]) into the mesh between the camshaft idler gear and the valve camshaft gear.

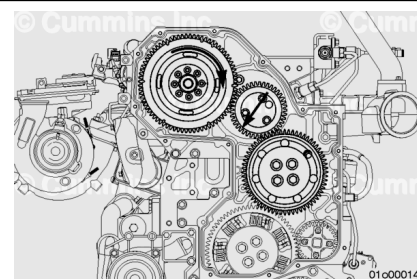
Note : Per the above adjustable idler gear install procedure, verify the oil hole in the retainer plate and the shaft are in line with the oil drilling in the head. Use the camshaft idler gear retainer markings to insert the feeler gauges in the correct teeth mesh. The triangles represent idler gear teeth. The lines represent the feeler gauge placement.



Shim Tolerance		
mm		in
0.0762	MIN	0.003
0.1016	MAX	0.004

Use hand pressure **only** to move the camshaft idler gear toward the center of the engine and rotate the cam gear **clockwise** (into mesh), so there is no gap on either side of the shim.

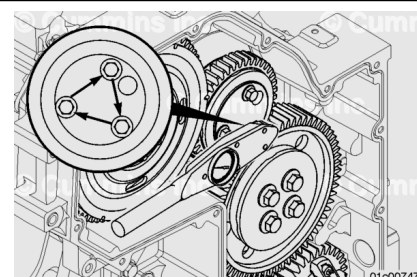
A slight drag will be felt on the shims while pulling the shims in and out. Do **not** remove the shims.



Hold the adjustable camshaft idler gear in position and tighten the capscrews. Use the alternating pattern shown in the illustration.

Tighten all capscrews in an alternating pattern.

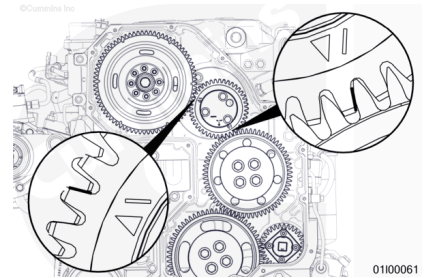
1. Tighten the camshaft idler gear mounting capscrews: 30 N•m [22 ft-lb].
2. Tighten the valve camshaft gear mounting capscrew: 30 N•m [22 ft-lb].
3. Rotate the camshaft idler gear mounting capscrews an additional 60 degrees.



4. Tighten the valve camshaft gear mounting capscrew: 102 N•m [75 ft-lb].
5. Rotate the valve camshaft gear mounting capscrew an additional 120 degrees.

Remove the shims.

A slight drag should be felt on the shims while pulling the shims out.



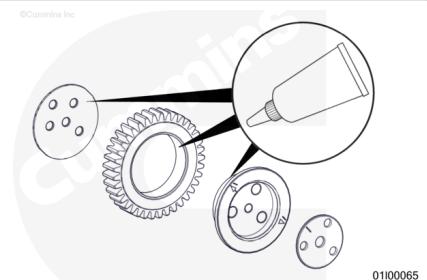
Fuel Pump Adjustable Idler Gear

Note : The adjustable fuel pump must be installed and timed correctly before installing the fuel pump idler gear.

Use the following procedure for fuel pump installation and timing. Refer to Procedure 005-016 in Section 5.

(/qs3/pubsys2/xml/en/procedures/377/377-005-016.html)

Apply Lubriplate™ to the thrust bearing plate, shaft, and fuel pump idler gear. Align all components in order shown in the illustration.



⚠ CAUTION ⚠

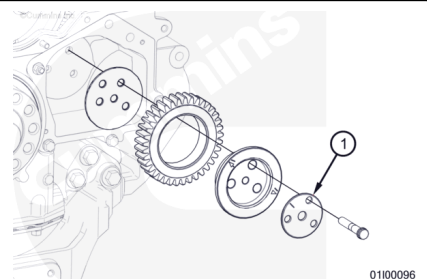
The adjustable fuel pump idler gear thrust bearing plate and shaft must be oriented such that the oil drain from the head is not blocked. If blocked, engine operation will be affected.

Replace the capscrews with new ones. Chase the internal threads of the block.

With new capscrews, verify the capscrews have pre-applied threadlocker or add threadlocker as specified.

If using plain capscrews, apply Loctite® 242, Part Number 3824040 or 3824041, to the threads of the fuel pump idler gear capscrews.

Install the adjustable fuel pump idler gear assembly to the cylinder block. The correct order from cylinder block out: thrust bearing plate (markings facing toward the gear), fuel pump idler gear, shaft, and retainer plate assembly.

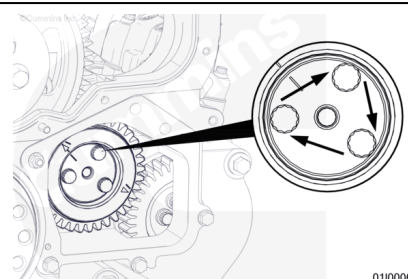


- The gear should be installed with the word "Pan" and arrow pointing toward the oil pan.
- The shaft **must** be installed with the alignment marks pointed toward the center of the lower idler gear and pointed toward the center of the fuel pump gear.
- Use the guide pins to start capscrews onto gear housing.

Note : This piece (1) is only on the adjustable idler gear.

⚠ CAUTION ⚠

The adjustable fuel pump idler gear retaining capscrew torque process must be completed within fifteen minutes of the initial threading or severe engine damage can occur. If disturbed after fifteen minutes, the capscrews must be cleaned and new medium strength threadlocker applied or use new pre-coated capscrews.

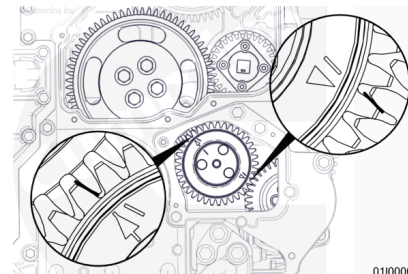


Install and tighten capscrews.

Hand tighten the capscrews. Loosen 60 degrees.

⚠ CAUTION ⚠

If the adjustable fuel pump idler gear mesh setting is beyond tolerances, it can lead to improper engine function or engine failure.



Insert narrow or tapered shims at the points illustrated:

- 0.1016 mm [0.004 in] into the gear teeth mesh between the lower idler gear and fuel pump idler gear
- 0.1016 mm [0.004 in] into the mesh between the fuel pump idler gear and fuel pump gear.

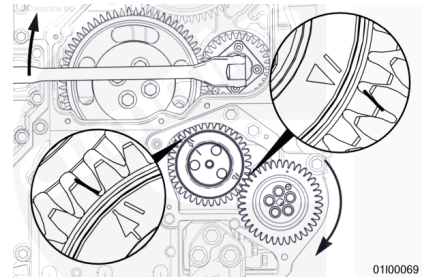
Note : Use the fuel pump idler gear shaft retainer markings as a guide to correct feeler gauge placement. The triangles represent idler gear teeth. The lines represent the feeler gauge.

Shim Tolerance		
mm		in
0.0762	MIN	0.003
0.1016	MAX	0.004

Move the fuel pump idler gear toward the center of the engine so there is no gap on either side of the shims.

Use a pry bar and the gear train housing to rotate the fuel pump gear **clockwise** and hold the fuel pump idler gear mesh in position.

Note : The fuel pump idler gear is in mesh when a slight drag can be felt on the shims while pulling the shims in and out. Do not remove the shims.

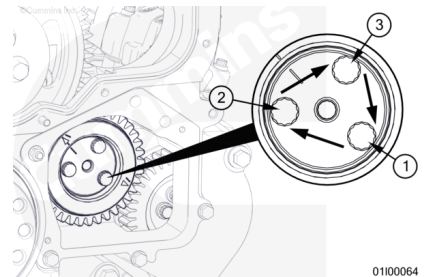


Hold the adjustable fuel pump idler gear in position. Tighten the capscrews. Use the pattern shown in the illustration.

Torque Value:

1. 35 n•m [26 ft-lb]
2. Rotate each capscrew an additional 60 degrees in the sequence shown.

Remove the shims.



Fuel Pump Non-Adjustable Idler Gear

Confirm the fuel pump is timed correctly before installing fuel pump idler gear. Refer to Procedure 005-016 in Section 5. (</qs3/pubsys2/xml/en/procedures/377/377-005-016.html>)

Apply Lubriplate™ to the thrust bearing, shaft, and fuel pump idler gear. Verify enough Lubriplate™ is used to hold thrust washer in place.

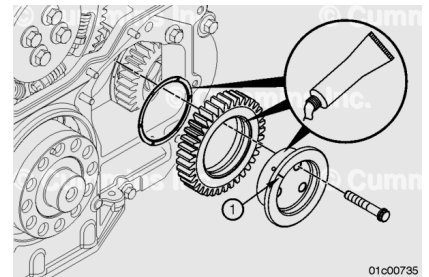
Remove any Loctite® residue. Chase the internal threads.

Install new capscrews.

With new capscrews, verify the capscrews have pre-applied threadlocker or add threadlocker as specified.

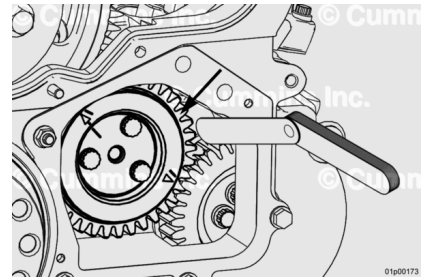
If using plain capscrews, apply Loctite® 242, Part Number 3824040 or 3824041, to the threads of the fuel pump idler gear capscrews.

Install the thrust bearing, shaft, and fuel pump idler gear. The shaft **must** be installed with the alignment mark (1) (stamped on the gear retainer) pointed toward the center of the lower idler scissor gear.



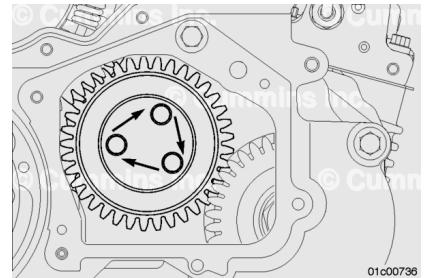
Use feeler gauge to verify 0.43 mm [.0169 in] in the location shown in the illustration.

If measurement between gear and idler hub is more than 0.43 mm [.0169 in] the gear **must** be removed and thrust washer installed correctly.



⚠ CAUTION ⚠

The fuel pump idler gear retaining capscrew torque sequence (including torque check) must be completed within 15 minutes of the initial threading or severe engine damage can occur. If disturbed after 15 minutes, the capscrews must be cleaned and new medium strength threadlocker applied, or use new pre-coated capscrews.



Install the capscrews.

Tighten all capscrews evenly.

Torque Value:

1. 35 n•m [26 ft-lb]
2. Rotate each capscrew an additional 60 degrees in the same sequence, as shown.

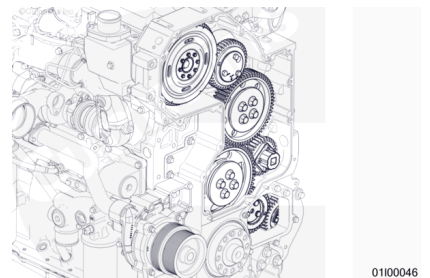
Do **not** load the lower idler scissor gear, if equipped, until all other gears are installed and tightened.

Load the lower idler scissor gear, if equipped, by tightening all of the gear adjusting screws.

Torque Value: 29 n•m [21 ft-lb]

⚠ CAUTION ⚠

Verify all gear screws are tightened to the proper torque so they do not back out during engine operation. Severe engine damage can occur.

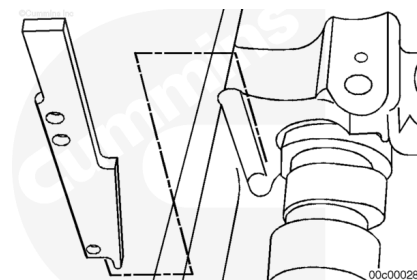


Note : If the fuel pump and air compressor have been removed, do not perform these last three steps until the fuel pump and air compressor have been installed.

Coat the entire gear train with clean 15W-40 engine oil.

Remove the valve camshaft timing wedge.

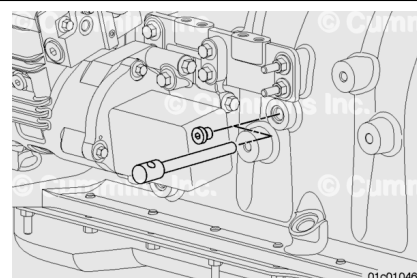
A sudden jerking motion upward is the most effective way to remove the wedge.



Remove the crankshaft locking pin.

Install the plug into the timing pin boss hole.

Torque Value: 20 n•m [177 in-lb]



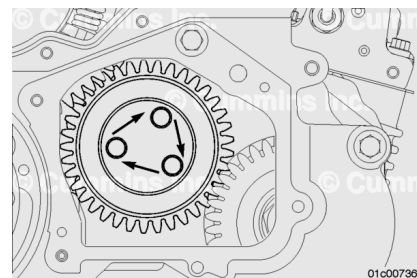
Rotation Check

Note : Recheck limit is **not** the total of both clearances.

Rotate the crankshaft over 360 degrees.

Using feeler gauges, check the gear lash between the following gears:

- Fuel pump idler gear to fuel pump gear
- Fuel pump idler gear to lower idler gear.



Mesh Tolerance		
mm		in
0.0508	MIN	0.002
0.1524	MAX	0.006

The mesh tolerance recheck limits in the table is **not** the total of both measurements.

Check the torque on the fuel pump idler gear capscrews.

Tighten the fuel pump idler gear capscrews.

Torque Value: 70 n•m [52 ft-lb]

If any of the capscrews rotate prior to achieving the above torque value:

- Loosen all three capscrews

- Tighten the capscrews. Use the method specified in the Install Step of this procedure for the fuel pump idler gear.

Finishing Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the camshaft position sensor. Refer to Procedure 019-363 in Section 19. (</qs3/pubsys2/xml/en/procedures/377/377-019-363.html>)
- Install the upper gear cover. Refer to Procedure 001-079 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-079.html>)
- Install the crankshaft position sensor. Refer to Procedure 019-365 in Section 19. (</qs3/pubsys2/xml/en/procedures/377/377-019-365.html>)
- Install the naturally aspirated air compressor plumbing and components, if equipped.
- Install the lower gear cover. Refer to Procedure 001-080 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-080.html>)
- Install the rocker lever cover and gasket. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/377/377-003-011-tr.html>)
- Install the EGR crossover tube. Refer to Procedure 011-070 in Section 11. (</qs3/pubsys2/xml/en/procedures/377/377-011-070-tr.html>)
- If coolant, oil, excessive fuel, or excessive black smoke has entered the exhaust system, the aftertreatment system **must** be inspected.
 - Use the following procedure to check the aftertreatment selective catalytic reduction. Refer to Procedure 014-025 in Section 14. (</qs3/pubsys2/xml/en/procedures/377/377-014-025.html>)
 - Use the following procedure to check the aftertreatment diesel particulate filter. Refer to Procedure 014-016 in Section 14. (</qs3/pubsys2/xml/en/procedures/493/493-014-016.html>)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to normal operating temperature. Check for leaks and proper operation.

